Hari Krishna Vydana

Tech Lead – Physical AI SDK | Agentic & Conversational AI

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Scientific Interests

Physical AI SDK, Edge AI Inference, Unified Inference Engines (CPU·GPU·NPU), Robotics Perception Pipelines, Model Optimization & Quantization (ROCm/RyzenAI), VLA/VLM Systems, Text and Voice Agents, Multi-Agentic workflows, Dialog Systems, Speech Recognition, Large Language Models (LLMs), Multimodal LLMs, Speech-Translation, Spoken/Written Language Assessment (Ed-Tech), Machine Translation, Natural Language Processing (NLP), Text-to-Speech (TTS), Knowledge Distillation, RLHF, DPO

Experience

Feb 2026 – Present **Tech Lead – Physical AI SDK (Senior Member of Technical Staff)**, *AMD*, Bangalore, India.
- ***Vision & Strategy*** – Conceived and owns the roadmap for **Physical AI SDK (PAVS)** — AMD's unified edge-inference platform (CPU·GPU·NPU) countering NVIDIA's NIM/Jetson lock-in; authored executive market analysis identifying AMD's **softpower gap**, framed the $47.5B Physical/Voice AI opportunity (35% CAGR), and drove internal alignment to resource the initiative.
- ***Team Leadership*** – Lead and mentor a cross-functional team of **9 engineers** across model inference, power & performance, SDK delivery, and customer co-engineering — setting technical direction, conducting design reviews, and driving quarterly deliverables from prototype to production.
- ***SDK Architecture*** – Designed the **unified inference engine**: single API across CPU, GPU, and NPU with multi-language bindings (Python, C++, Kotlin, JavaScript) and zero-copy edge transport — reducing developer time-to-first-inference from weeks to a day.
- ***Technical Execution*** – Directed **PaDiM** optimization (1.6 → 125 FPS, **77**×, zero accuracy loss); led **SmolVLA** VLA tuning (**37% latency reduction**, 48 → 30 ms on ROCm 7.2.4); operationalized MobileSAM, YOLOv12, YOLO26, CenterPoint into AARTS robotics pipelines with MIGraphX (**2**× over PyTorch+ROCm).
- ***Ecosystem & Customers*** – Defined model zoo strategy (research tarball → off-the-shelf packaged models per device); delivered ROCm + RyzenAI installer, vLLM/Llama.cpp/FastFlowLM stack, and Yocto/LTS edge support; initiated co-engineering with **Rivian** (in-car monitoring & voice assistant) and **Google** (Gemma on LiteRT on AMD hardware).

Dec 2025 – July 2025 **AI Engineer**, *Complex Chaos (Through Rippling)*, Zurich, Switzerland.
- ***Alignment.AI*** – Developed AI systems for Aligning Human opinions At Scale.
- ***Alignment.AI*** – Led the design and deployment of LLM-Agents for Strategy-Negotiations and Consensus building.

June 2025 – Dec 2024 **Tech Lead – Voice AI**, *Telepathy AI Labs GMBH*, Zurich, Switzerland.
- ***Appointment.AI*** – Developed scalable, goal-oriented multi-agent dialog systems (text/voice) appointment booking and task management.
- ***Appointment.AI*** – Led the design and deployment of LLM-driven agents for automated appointment scheduling with adaptive, real-time dialog capabilities .
- ***ASR & Dialog Management*** – Engineered ASR, VAD, and dialog turn detection for real-time voice interactions.
- ***Agentic AI & LLMs*** – Built an LLM-driven Multi-Agentic systems for automated appointment scheduling.
- ***TTS Optimization*** – Integrated cost-effective TTS models, optimizing latency and inference costs.

Nov 2024 – Oct 2023 **Research Associate – ALTA Project**, *University of Cambridge, UK*.
- ***ALTA Project*** - Contributed to Automated Language Teaching and Assessment (ALTA).
- Developed ***Automatic evaluation of spoken summaries using Large Language Models (LLMs)***.
- Researched ***Automatic spoken language assessment with Human-in-the-Loop validation***.
- ***Distillation:*** – Distilled audio foundation models to optimize inference compute.
- ***LLM's For Analytic Scoring:*** – Proposed analytic scoring methods and benchmarked on spoken assessment.
- ***Multi-View Error Mitigation:*** – Leveraged LLMs for multi-view error correction in language assessment.

Sept 2023 –Nov 2022 **Senior Researcher**, *Cerence, Aachen, Germany*.
- Developed voice-enabled digital assistants for real-time interactions in Advanced Driver Assistance Systems.
- Developed ***On-Device and Cloud Speech Recognition models*** for automotive applications.
- Worked on ***Training/Evaluating/Benchmarking*** ASR models targeted for Automotive Domains.

Oct 2022 – Sept 2021 **Senior ASR Researcher**, *Huawei Technologies Oy (Through GITPS consultant), R&D Center, Finland*.
- Built ***Voice Search models*** for ***Petal Search*** in Huawei devices.
- Developed and Deployed Speech and Language Models for **Voice operated Web Search**.
- Designed ***On-Device ASR models*** for Huawei mobiles.

Education

July 2021 – Jan 2019 **Postdoctoral Researcher**, *Brno University of Technology (BUT), Brno, Czech Republic*.
Research Focus: Speech Translation Systems, End-to-End Speech Recognition
Advisors: Doc. Ing. Lukáš Burget & Prof. Doc. Ing. Jan Černocký

Dec 2018 – Dec 2014 **Ph.D. in Multilingual Speech Recognition**, *III-T Hyderabad, India*.
Thesis: Multilingual Speech Recognition for Indian Scenarios
CGPA: 9.5/10 **Advisor:** Dr. Anil Kumar Vuppala

Research Projects

- **LLMs for Managing Conversations**

  " LLMs for Conversation Generation and Assesment."

  - **Voice-operated appointment booking and service systems** with multi-agent frameworks for dynamic conversations.
  - **Stefano Bano, Hari Krishna Vydana, Kate Knill, Mark J.F. Gales.**
    "Can GPT-4 Perform Analytic Assessment of L2 Writing?" In Proc. Workshop on Innovative Use of NLP for Building Educational Applications, ACL 2024.
  - Finetunig LLMs with parameter efficient training, LORA, QLORA
  - Preference training LLMS with RLHF (Reinforcement Learning with Human Feedback) and DPO(Direct Parameter Optimization).
  - Developed **multi-view error mitigation systems** (LLMs), **neural text graders** (BERT, RoBERTa models) for language assessment.

- **Transformer-based Speech Translation Systems**

  "Explored approaches to integrate ASR with NLP modules, developing models with an end-to-end differentiable path between ASR and NLP modules (MT). These models reduce error propagation across the pipeline."

  - Developed Cascaded, End-to-End systems, and Jointly-optimized ASR-MT systems.
  - **Hari Krishna Vydana**, Martin Karafi'at, Lukáš Burget, Honza Černocký.
    "The IWSLT 2021 BUT Speech Translation Systems", Proceedings of the 18th International Conference on Spoken Language Translation (IWSLT 2021).
  - **Hari Krishna Vydana**, Martin Karafi'at, Katerina Zmolíková, Lukáš Burget, Honza Černocký.
    "Jointly Trained Transformer Models for Spoken Language Translation", Proc. ICASSP, Toronto, Canada, 2021.

- **End-to-End Automatic Speech Recognition Systems**

  - **Speech Recognition for Indian Scenarios:**

    "Developed ASR systems for code-mixing/code-switching environments. Proposed a model that handles recognition and text rendering separately to operate in code-mixed contexts."

    - **Hari Krishna Vydana**, Krishna Gurugubelli, Vishnu Vidyadhara Raju V, and Anil Kumar Vuppala. ***"An Exploration Towards Joint Acoustic Modeling for Indian Languages"***, Interspeech 2018, Hyderabad, India.
    - **Thesis:** ***"Salient Features for Multilingual Speech Recognition in Indian Scenarios"***.

  - **Combining Hybrid and End-to-End ASR Models:**

    "Trained End-to-End ASR systems and combined them with hybrid models to enhance performance."

    - Martin Karafiát, Murali Karthick Baskar, Igor Szöke, **Hari Krishna Vydana**, Karel Veselý, and Jan Černocký. ***"BUT Opensat 2019 Speech Recognition System"***, arXiv:2001.11360, 2020.
    - Katerina Zmolikova, Martin Kocour, Federico Landini, Karel Beneš, Martin Karafiát, **Hari Krishna Vydana**, Alicia Lozano-Diez, Oldrich Plchot, and Murali Karthick Baskar. ***"BUT System for CHiME-6 Challenge"***, ICASSP 2020, Barcelona, Spain.
    - **Hari Krishna Vydana**, Sivanand Achanta, and Anil Kumar Vuppala. ***"Incorporating Speaker Normalization to End-to-End ASR"***, SLTU 2018, India.
    - Ekaterina Egorova, **Hari Krishna Vydana**, Lukáš Burget, and Jan Černocký. ***"Out-of-Vocabulary Words Detection in End-to-End ASR"***, Interspeech 2021, Brno, Czech Republic.
    - Ekaterina Egorova, **Hari Krishna Vydana**, Lukáš Burget, and Jan Černocký. ***"Spelling-Aware Word-Based End-to-End ASR"***, IEEE Signal Processing Letters, 2022.

Evaluations participated & Open-source Repositories

- **IWSLT-2020, IWSLT-2021**: Offline speech Translation
- ASR Challenge for Indian English-2021: First place
- **OPENSAT-2019 & OPENSAT-2020** : First place in ASR
- CHiME-6 Workshop (satellite of ICASSP 2020)
- Low Resource Speech Recognition Challenge for Indian Languages (Interspeech 2018 Special Session)

Open-source Repositories.

- Transformer ASR:
- Transformer-Machine Translation:
- End-to-End ASR:
- Jointly-Trained Transformers for Spoken Language Translation:

Skills

ML	Agents-SDK, LangChain, LangGraph, Semantic Kernel, PyTorch, Numpy, Scipy, Pandas, Matplotlib, PyTorch Lightening, Databricks, Pipecat, LiveKit, Retell, WebRTC, Websocket.
Speech	KALDI, EESEN, ESPNET, FAIRSEQ, NEMO, WENET, Huggingface, AZURE, OPENAI, FASTAPI, Azure Graphs, Authentications.
Data-Tools	SQL, Mongo-DB, Pydantic, Redis, PySpark.
UX-UI	REACT, Node.js, STREAMLIT, TWILLIO

Publications

Link for the complete list of papers:, *Google scholar*.